NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR303575
Lead Applicant	Mr James Alexander
Research Title	A Hybrid delivery of exercise for physical self-management in people with Parkinson's Disease – a single-arm feasibility study with pre and post-test design.
Plain English Summary	**Aim(s) of the research** I was inspired from working with people with Parkinson's disease how exercise can help. This project will explore if hybrid exercise groups help people with Parkinson's disease to exercise. A hybrid exercise group allows people to attend the group together with other people or online from home. Findings could help us support more people with Parkinson's disease to exercise regularly. **Background to the research** Parkinson's disease affects the body's nervous system. It causes people to feel stiff, shake and difficulty walking and keeping their balance. It also causes people to feel sad and anxious. These will worsen over time. Exercise is helpful, but people with Parkinson's disease exercise less than people without it. They need support to exercise regularly, especially if they have trouble walking or don't live near to an exercise group. An online exercise group means people can exercise at home. Researchers say online exercise groups are safe and can help people with Parkinson's disease feel better. But exercising online makes it difficult to talk to and see each other. A hybrid exercise group allows people choice to attend together with other people or online from home. This might help people with Parkinson's disease exercise more. We need to see if this is true and how it might work. **Design and methods used** The project will investigate two questions: a. Look at existing research on this topic through a scoping review. b. Find out if it's possible to run a hybrid exercise group research study. The scoping review will find research on different types of online exercise groups for people with Parkinson's disease and how they run. All the online exercise groups will be looked at individually. Similarities and differences between them can be identified. I can then compare different exercise groups against each other in the future. The hybrid exercise group is created for this project. I need to find out if it works and if a future large study investigating the group is possible. This means if the hybrid exercise group is carried out as planned, if people want to take part in the project, if those that do like it or not and if it helps participants to exercise. The exercises in this hybrid group are for people with Parkinson's disease who have less difficulty with their walking and balance. Participants will attend the group 1x per week for 12 weeks. It will include 60-minutes of exercise and 30-minutes afterwards for participants to talk together about how to exercise more regularly. Participants can attend in-person together or online from home. I will monitor the group to see how well it is carried out and ask participants for their feedback on the project. **Patient and public involvement** I worked with people with Parkinson's disease and their family members. They agreed on the project idea, importance of making the exercise group accessible and helped design the hybrid exercise group. We will continue working together to finalise the exercises. They will also help write easy-to-read documents for the project. **Impact & Dissemination** This study will help understand if an on online and hybrid exercise group for people with Parkinson's disease is needed and if so, guide a future larger study. This will provide a way for people with Parkinson's disease to be more physically active which is beneficial for staying healthy and managing their symptoms. The findings will be written in scientific journals. With help from Parkinson's UK charity, I will share project findings with participants from the study, other people with Parkinson's disease and their families and healthcare professionals through talks and information sheets.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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1. Application Summary Information

Host organisation (which will administer any award)
University College London

Partner organisation	Central and North West London NHS Foundation Trust

Research Title	A Hybrid delivery of exercise for physical self-management in people with Parkinson's Disease – a single-arm feasibility study with pre and post-test design.

Partnership Fellowship	N/A

Proposed WTE Please select which WTE option you would prefer
1.00

Duration (months) The duration for your award is based on your chosen WTE
36

Proposed start date if grant awarded	01/04/2024

Research Type	Feasibility study, Evidence Synthesis

2. Applicant CV

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
09/2011	07/2014	MSc	Advancing Spinal Cord Injuries Rehabilitation & Management	United Kingdom	Buckinghamshire New University	Pass (63%)
09/2004	07/2007	BSc(Hons)	Physiotherapy	United Kingdom	University of the West of England, Bristol	2:1

Present and previous positions

(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
09/2018	-	Band 7 Physiotherapist (0.6FTE)	Central & North West London NHS Foundation Trust	0
02/2018	-	Senior Lecturer (0.4FTE)	St George's University, London	0
02/2018	09/2018	Band 6 Physiotherapist	Central & North West London NHS Foundation Trust	0
03/2015	09/2017	Teaching Fellow	King's College London	0
04/2011	12/2014	Band 6 Physiotherapist	Buckinghamshire Healthcare NHS Trust	0
12/2007	03/2011	Band 5 Physiotherapist	Salisbury NHS Foundation Trust	0
10/2007	12/2007	Band 5 Physiotherapist	United Lincolnshire Hospitals NHS Trust	0

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
CNWL research starter grant	Lead investigator and applicant for the grant. Developed research protocol for study to explore experiences of online exercise groups for people with Parkinson's disease. Used also to develop NIHR PCAF application.	8875	Central & North West London NHS Foundation Trust	6	03/08/2020	05/02/2021
KCL/Wellcome Trust public engagement small grants scheme award	Lead applicant	750	KCL/Wellcome trust	12	01/05/2017	30/04/2018

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Alushi, L., Alexander, J., Jones, J. and Lafortune, L.. (2022) ‘A Systematic Review on Physical Health Education Interventions for People with Parkinson's Disease: Content, Impact, and Implementation Considerations Across

Publication Record

This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
the Parkinson's Trajectory.’, *Journal of Parkinson's disease*, 12(5).
Sheehan, KJ., Guerrero, EM., Tainter, D., Dial, B., Milton-Cole, R., Blair, JA., Alexander, J., Swamy, P., Kuramoto, L., Guy, P., Bettger, JP. and Sobolev, B.. (2019) ‘Prognostic factors of in-hospital complications after hip fracture surgery: a scoping review.’, *Osteoporosis international : a journal established as result of cooperation between the European Foundation for Osteoporosis and the National Osteoporosis Foundation of the USA*, 30(7).
Alexander. (2019) ‘Technology-enhanced learning in physiotherapy education: Student satisfaction and knowledge acquisition of entry-level students in the United Kingdom’, *Research in Learning Technology*, 27.
Sheehan, KJ., Williamson, L., Alexander, J., Filliter, C., Sobolev, B., Guy, P., Bearne, LM. and Sackley, C.. (2018) ‘Prognostic factors of functional outcome after hip fracture surgery: a systematic review.’, *Age and Ageing*, 47(5).
Bazin. (2016) ‘Learning with expert patients – can this support development of clinical skills?’, *Physiotherapy (United Kingdom)*, 1(102).

Relevant Prizes, Awards and other Academic Distinctions

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/07/2016	Higher Education Academy	Fellow	

ORCID iD 0000-0002-9459-7228

3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Physiotherapist

Do you have current / active professional registration?
This registration must be with one of the **ICA approved regulatory bodies**

Yes

Professional registration number/pin
Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)

PH81921 - HCPC

Please describe your research career to date

I'm a clinical academic physiotherapist working as a Senior Lecturer at St George's University, London (SGUL) and as a specialist neurological physiotherapist for the Camden Neurology & Stroke Service (CNSS), an NHS community neurological rehabilitation service. My clinical academic career interest began during my MSc and I have continued exploring allied health professional (AHP) clinical academic roles during my PCAF and Post-Graduate Certificate in Healthcare Research Skills and Methods and will continue to develop as one through completion of my considered DCAF plan. My ambition is to progress my clinical academic career within the NHS supported by an HEI with excellent clinical and research credentials in neurology and neurorehabilitation. This will enable me to be a leading clinical academic in the field of exercise for people with Parkinson's Disease (PD), inspiring others to positively impact on patients' lives. As a successful clinical academic I will continue demonstrating excellent research skills, becoming a world-leading healthcare researcher and advocate for AHP research. I believe my research and clinical experiences to date provide strong supporting evidence for this DCAF.

Clinical experience

I have 10 years' experience in neurological rehabilitation and belong to special interest groups including the Association of Chartered Physiotherapists in Neurology and the Parkinson's Excellence Network exercise hub. Since 2018 I have worked for the CNSS supporting patients with neurological conditions. I have excellent clinical skills including rehabilitation and disease management for complex neurological conditions including PD. I have developed specialist clinical skills to support people with PD (PwP) through exercise by completing PD Warrior™ training and deepened my understanding of complex PD management through attending the 3rd Queen Square Movement Disorders short course, led by world-leading movement disorders specialists. These skills I have imparted to clinical teams within CNWL trust through regular teaching and joint patient visits to enhance evidence-based care for PwP. I will continue developing specialist skills to become a leading clinical academic through my training and development plan. I have strong leadership skills, developed through completing the NHS Leadership Academy Edward Jenner course, supporting innovation within my team to refine our provision of exercise groups for our patients. I supported a service evaluation of our exercise group for people with neurological conditions alongside my physiotherapy lead, which she presented at the 2023 Congress on Neurorehabilitation and Neural Repair in Maastricht. Inspired by emerging evidence for exercise for PwP, I wanted to address the gap in service provision for this population. Embedding evidence-based research, clinical skills and utilising the Quality Improvement approach I developed an exercise and self-management group. This was achieved through logic modelling, stakeholder involvement, clinical skills training (PD warrior), data collection and evaluation. During the COVID-19 pandemic I transferred this to an online format. This alongside the NHS Long-term Plan, which advocates greater use of e-health to provide equity of access to healthcare for those with long-term conditions and underserved populations, has sparked my desire to research hybrid formats of group exercise for PwP.

Research experience

I completed my MSc in 2014 and will complete my PG Certificate in Research methods in June 2023. These

courses provided an excellent foundation for critically understanding and evaluating clinical research, its design, conduct, dissemination and project management and appropriate statistical methods. They, alongside GCP training, have equipped me with the knowledge and depth of understanding of research governance and organisation to embark on a DCAF.

Research projects

I have been involved with independently designing, conducting, managing and publishing several research projects, as follows:

Clinical research:

In developing my DCAF research topic I have successfully led two research projects during my PCAF. These explored PwP and exercise professionals' experiences, respectively, of online exercise groups using a phenomenological approach through semi-structured interviews. These projects, together with focussed training courses, have given me the ability to formulate a research question, design a methodologically sound study, obtain ethical approval, undertake recruitment, data collection and analysis and publication. Despite setbacks in carrying out these studies due to COVID-19, the first study has been accepted for poster presentation at the World Parkinson's Congress in Barcelona in July 2023, at Physiotherapy UK conference in November 2023. It has been submitted for publication in Disability and Health Journal. The latter study will be written up for publication by December 2023 and I will present findings at the SGUL Research Day in December 2023.

During my PCAF I collaborated with a PhD student and co-investigated a systematic review into physical activity education for PwP. My role included title and abstract screening, full-text screening, data collection, quality appraisal and contributing to the reporting. This study was published in the renowned Journal of Parkinson's Disease.

I have supervised 4 undergraduate and 2 post-graduate physiotherapy student research projects. Projects have included cross-sectional and mixed methods study design and focused on physiotherapy provision for early-stage PD and student experiences of physical activity guideline promotion for neurological patients.

Educational research:

I was chief investigator for a mixed methods study exploring technology-enhanced learning in undergraduate physiotherapy education. The study utilised a survey and focus groups to explore student experiences whilst also comparing grade attainment. I was responsible for study design, survey development, data collection and analysis. I presented this study via oral presentation at the World Congress of Physical Therapy and published it in Research in Learning Technology journal.

Relevant experience

I have 8 years' experience of teaching in higher education. Working within research institutes allows me to collaborate with other research-oriented AHPs. My extensive knowledge of pedagogy and teaching methods contributed to obtaining Fellowship of the Higher Education Academy. I recognise during my DCAF and clinical academic career I will have a responsibility to teach clinical and research skills and explain the importance of these to our practice. I believe I have the skills to do this.

My interest in technology for people living with neurological conditions led me to join a steering group for a project exploring assistive technology to empower future care workforces. This has given me skills in engaging and utilising steering groups to shape and design research topics.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
No

Mitigating Factors
Please use this question to detail any mitigating factors you wish to make the Selection Committee aware of. The Applicant Guidance Notes provide guidance regarding completion of this question.

None to declare

4. Plain English Summary of Research

Aim(s) of the research

I was inspired from working with people with Parkinson's disease how exercise can help. This project will explore if hybrid exercise groups help people with Parkinson's disease to exercise. A hybrid exercise group allows people to attend the group together with other people or online from home. Findings could help us support more people with Parkinson's disease to exercise regularly.

Background to the research

Parkinson's disease affects the body's nervous system. It causes people to feel stiff, shake and difficulty walking and keeping their balance. It also causes people to feel sad and anxious. These will worsen over time.

Exercise is helpful, but people with Parkinson's disease exercise less than people without it. They need support to exercise regularly, especially if they have trouble walking or don't live near to an exercise group. An online exercise group means people can exercise at home.

Researchers say online exercise groups are safe and can help people with Parkinson's disease feel better. But exercising online makes it difficult to talk to and see each other. A hybrid exercise group allows people choice to attend together with other people or online from home. This might help people with Parkinson's disease exercise more. We need to see if this is true and how it might work.

Design and methods used

The project will investigate two questions:

a. Look at existing research on this topic through a scoping review.
b. Find out if it's possible to run a hybrid exercise group research study.

The scoping review will find research on different types of online exercise groups for people with Parkinson's disease and how they run. All the online exercise groups will be looked at individually. Similarities and differences between them can be identified. I can then compare different exercise groups against each other in the future.

The hybrid exercise group is created for this project. I need to find out if it works and if a future large study investigating the group is possible. This means if the hybrid exercise group is carried out as planned, if people want to take part in the project, if those that do like it or not and if it helps participants to exercise.

The exercises in this hybrid group are for people with Parkinson's disease who have less difficulty with their walking and balance. Participants will attend the group 1x per week for 12 weeks. It will include 60-minutes of exercise and 30-minutes afterwards for participants to talk together about how to exercise more regularly. Participants can attend in-person together or online from home. I will monitor the group to see how well it is carried out and ask participants for their feedback on the project.

Patient and public involvement

I worked with people with Parkinson's disease and their family members. They agreed on the project idea, importance of making the exercise group accessible and helped design the hybrid exercise group. We will continue working together to finalise the exercises. They will also help write easy-to-read documents for the project.

Impact & Dissemination

This study will help understand if an on online and hybrid exercise group for people with Parkinson's disease is

needed and if so, guide a future larger study. This will provide a way for people with Parkinson's disease to be more physically active which is beneficial for staying healthy and managing their symptoms.

The findings will be written in scientific journals. With help from Parkinson's UK charity, I will share project findings with participants from the study, other people with Parkinson's disease and their families and healthcare professionals through talks and information sheets.

5. Scientific Abstract

Background

Parkinson's disease (PD) is the fastest growing neurological disease in the world. Exercise groups are beneficial for supporting sustained physical activity habits in people with PD (PwP). Due to the COVID-19 pandemic and wider accessibility issues, PwP have utilised online exercise groups. Hybrid formats of group exercise, the flexibility of attending a class in-person or remotely online, may improve equity of access to exercise, supporting sustained physical activity habits. Little is known about online and hybrid group exercise for PwP and feasibility of a future clinical trial.

Research question

Can a hybrid format of group exercise support people with Parkinson's Disease (PD) to sustain physical activity habits?

Aim

Explore the feasibility of a Randomised Controlled Trial (RCT) of a hybrid format of group exercise to support sustained physical activity habits in people with PD.

Objectives:

1. Understand the existing evidence base on online and hybrid formats of group exercise for people with PD.

a) Understand the operational definitions used pertaining to hybrid and online exercise groups for people with PD.

b) Understand the components of online and hybrid exercise groups for people with PD.

c) Identify outcomes and impacts of online and hybrid exercise group for people with PD.

2. Determine the feasibility of an RCT of a hybrid format of group exercise to support sustained physical activity habits in people with PD.

Methods

WP1 – To continue developing the intervention a scoping review will address objectives 1 a-c. It will follow the Joanna Briggs Institute methodology for scoping reviews. It will include a three-step search strategy. 3 researchers will independently assess titles and abstracts, then review the full text of the selected articles, applying the inclusion criteria. Data will be extracted and will be presented in tables and summarised narratively following critical appraisal.

WP2 – A feasibility study will assess the acceptability, fidelity, and feasibility of a future RCT (objective 2). This will be a non-randomised, single-arm study with pre and post-test design using mixed methods. Convenience sampling will recruit 25-50 participants who will complete a 12-week exercise and peer self-management intervention.

Timeline for delivery

Year 1: PPI activities, WP1 protocol registration and publication, data collection & analysis, ethical approval for WP2, site set-up and training for WP2, WP2 protocol publication.

Year 2: PPI activities, WP1 outputs & dissemination, WP2 implementation, data collection, data analysis, ongoing training and support for practitioners delivering intervention.

Year 3: PPI activities, WP2 data collection, data analysis, WP2 feedback, outputs & dissemination, thesis completion, thesis dissemination, RCT protocol development.

Impact and dissemination

Findings will build on the evidence-base for online and hybrid group exercise for PwP and inform a future RCT which will lead to patient benefits. Our proposed theory of change is that small exercise habits developed in the

hybrid exercise groups will increase self-efficacy (through goal mastery and vicarious experience) and initiate and maintain behaviours which sustain activity. Findings will be shared through publication in academic journals, conference presentations and talks to PwP and their family through the Parkinson's UK charity.

6. Detailed Research Plan

What is the problem being addressed?

The Parkinson's Pandemic

Parkinson's Disease (PD) is a movement disorder and the second most prevalent neurodegenerative disease after Alzheimer's disease.[1,2] It causes motor and non-motor impairments including rigidity, bradykinesia, tremor and depression.[3-5] PD is the fastest growing neurological condition in the world and the global population of PD could exceed 17 million by 2040.[6] Its UK annual incidence is 15 to 20 per 100,000.[7]

Financial & Health-related impact of PD

Health & care costs are higher for people with PD (PwP) compared to the general population. Average annual costs are £2471 higher per patient at diagnosis increasing to £4004 per patient 10-years following diagnosis.[8] A study commissioned by Parkinson's UK identified PwP were £16,500 worse off per year due to lost income from an inability to work and costs of associated disability.[9] Compared to the general population PwP are typically one-third less active, attributed to reduced physical abilities, self-efficacy and mental health impairments.[5,10,11] Disease progression and reduced physical activity (PA) worsen motor symptoms leading to falls, fractures and cardiovascular co-morbidities causing increased hospital admissions.[12] Over 4 years, 232,905 non-elective admissions in England cost the NHS over £777 million.[13] Exercise is a promising, cost-effective, and low-risk intervention to improve motor and non-motor symptoms in PwP.[14]

Why is this research important?

Exercise for PD

Exercise is planned, structured and purposeful actions that aim to maintain or improve at least one component of physical fitness.[15] Exercise is a component of PA which describes movements of the body by skeletal muscles that requires energy expenditure.[15] A 2023 Cochrane review recommended exercise for managing PD symptoms.[16] Exercise improves gait,[17] strength,[18] balance,[19] physical activity,[20] mood and quality of life[21,22] in PwP. European and NICE guidelines for PD recommend physiotherapy to support exercise and PA, however no standardisation of care means variation in provision of exercise for PwP from NHS services.[5,7] Due to the multiplicity of PD symptoms no one type of exercise is recommended over another and exercise can be performed individually, in a group, supervised or unsupervised.[5] Whilst in-person exercise is popular, accessibility is difficult for people with transport or mobility issues or people in isolated communities.[23] Online exercise can reduce these barriers offering a more inclusive approach to being physically active.[24]

Sustaining PA in PwP is challenging. Automaticity in performing a new habit can take 10 weeks to develop.[25] Exercise interventions alone are insufficient for developing habits to support sustained PA. Self-management skills are integral to promote self-efficacy and behaviour change.[26] Exercise alongside self-management strategies for PwP could support self-efficacy to sustain PA.[27,28] There is an opportunity to establish exercise programmes within NHS services that are sufficient to illicit and sustain PA habits in PwP.[29]

Why is this research needed now?

The evidence gap.

Exercise for managing PD symptoms, including reduced balance and falls, is the top research priority as identified by the James Lind Alliance. Considering the growing PD population, financial costs associated with disease progression and physical inactivity and knowing the benefits of exercise for PwP there is a need to improve its accessibility. The COVID-19 pandemic demonstrated exercise groups can be delivered online. Studies have demonstrated online group exercise for PwP is feasible and could promote physical health benefits but this has not yet been supported in clinical trials.[30-32] During my PCAF I conducted research identifying the experiences of PwP and exercise professionals participating in online exercise groups. Findings demonstrated PwP welcome the flexibility to attend online should it be difficult to attend in-person.[33] Barriers to online exercise exist, including accessibility and technology illiteracy.[33] My research identified a preference for hybrid approaches, the flexibility to attend an exercise group either in-person or remotely online. This could improve equity of access to exercise, allowing PwP to sustain PA habits. There is limited understanding of the evidence supporting online and hybrid formats of groups exercise for PwP and no original research exploring hybrid formats of group exercise for PwP exists. It is unknown whether a Randomised Controlled Trial (RCT) exploring a hybrid format of group exercise is feasible.

What is the research question / aims and objectives?

Research question

Is a hybrid group exercise intervention for people with Parkinson's Disease (PD) feasible and acceptable?

Aims and objectives.

To explore the feasibility of an RCT of a hybrid format of group exercise for people with PD to support sustained exercise habits.

People with PD who were part of my PPI group revealed a need to understand the evidence base for online and hybrid exercise groups and wanted to know how a hybrid group would improve accessibility for all socioeconomic backgrounds. In response to this, objectives were identified to continue the development of this complex intervention in preparation for an RCT. This project utilises MRC guidance on developing and evaluating complex interventions and builds on developmental work completed during my PCAF.[34]

Objectives:

1. To understand the existing evidence base on hybrid and online formats of group exercise for people with PD.

a) To understand the operational definitions used pertaining to hybrid and online exercise groups for people with PD.

b) To understand the components of online and hybrid exercise groups for people with PD.

c) To identify the outcomes and impacts of online and hybrid exercise group for people with PD.

2. To determine the feasibility of an RCT of a hybrid format of group exercise to support sustained PA habits in people with PD.

Project Plan

Work package 1 (WP1) – Scoping review

A scoping review is chosen to map key concepts underpinning the topic and clarify working definitions, and the conceptual boundaries.[35] During my PCAF a systematic review was planned, however following an initial literature search there was a paucity of published studies exploring online and hybrid exercise groups for PwP. Discussions with the PPI group and project supervisors concluded a scoping review as the preferred methodology at present. Due to the emerging nature and limited published literature on hybrid formats of group exercise the review will include all formats of online group exercise, including hybrid formats.

Objective: to identify and understand the existing evidence base supporting online and hybrid formats of group exercise for PwP (objectives 1 a-c). This will support development of a theory of change and refine the hybrid exercise intervention in preparation for a feasibility study and future RCT.

Scoping review questions:

1. What are the operational definitions used for online and hybrid exercise group studies?

2. What study methods have been used to investigate online and hybrid exercise group interventions?

3. What are the common intervention components of online and hybrid exercise groups?

4. What outcome measures are used to assess online and hybrid exercise groups?

Participants

We will include studies which target people with idiopathic PD at all stages (Hoehn & Yahr stages 1-5) of all ages. Studies specifically looking at people with atypical Parkinson's will be excluded as exercise approaches for these conditions differ compared with idiopathic PD.

Concept

This review will focus on studies evaluating exercise delivered through an online group format specifically for PwP. A group is defined as an intervention involving two or more participants. The intervention can include any form of exercise/PA. Included studies will need to have delivered a "live online" exercise group i.e., not solely pre-recorded, and deliver an exercise/PA component with or without other components e.g., education or peer support.

Context

Studies will be included if the intervention took place remotely using an online platform. This could include online video conferencing platforms e.g., Zoom or MS Teams, other digital platforms and mobile apps. Studies which have incorporated a hybrid approach will be included.

Types of Sources

This review will consider experimental and quasi-experimental study designs including RCTs, non-randomized controlled trials, before and after studies, interrupted time-series studies, piloting and feasibility studies. Due to the novel nature of eHealth interventions, studies without control groups will be included. Additionally, analytical observational studies including prospective and retrospective cohort studies, case-control studies and analytical cross-sectional studies will be considered for inclusion. We will consider descriptive observational study designs including case series, individual case reports and descriptive cross-sectional studies for inclusion. Qualitative studies will be considered that focus on, but not limited to, designs such as phenomenology, grounded theory, ethnography, qualitative description, and action research. Additionally, systematic reviews that meet the inclusion criteria will be considered, depending on the research question. Finally, text and opinion papers will be considered. Only studies published in English will be included. There will be no limitation on publication date.

Methods

This review will be conducted in accordance with the Joanna Briggs Institute (JBI) methodology for scoping reviews.[36]

Search strategy

The search strategy utilises the JBI 3 step methodology. It will identify published and unpublished studies. An initial limited search of MEDLINE, CINAHL and Embase will identify articles on the topic. This preliminary search will use these selected keywords: (Parkinson's disease or Parkinson's) AND ((online exercise) OR (virtual exercise) OR (remote exercise) OR (e-health) OR (home-based exercise) OR (telehealth) OR (telemedicine)) AND ((Exercise OR (physical activity)). Words contained in the titles and abstracts of relevant articles and the index terms used to describe the articles will be used to develop a full search strategy for selected databases: MEDLINE/PubMed, CINAHL, Web of Science, Embase, Cochrane Library and PsychInfo. The search strategy will be adapted for each included database and/or information source. The search strategy will be an iterative process and may be adapted as reviewers become familiar with the evidence base.[36] The reference list of included sources will be screened for additional studies. Sources of unpublished studies/ grey literature to be searched include Google, Web of Science, Embase and Clinical Trials Register.

Study/Source of Evidence selection

All identified citations will be collated, uploaded into EndNote and duplicates removed. Following a pilot test, titles and abstracts will be screened by two or more independent reviewers against the inclusion criteria. Potentially relevant sources will be retrieved in full, and their citations imported into EndNote. Full texts of selected citations will be assessed against the inclusion criteria by two or more independent reviewers. Reasons for exclusion at full text will be recorded and reported in the scoping review. Disagreements between reviewers during the selection process will be resolved through discussion or with an additional reviewer/s.

Data Extraction

Data will be extracted by two or more independent reviewers using a data extraction tool developed by the reviewers, informed by JBI data extraction recommendations[37] but modified to include all pertinent information

relevant to the review aims and objectives. The tool will be pilot tested by two researchers by extracting data from 3 papers and comparing results. The data extraction tool will be modified as necessary during data extraction.

Data Analysis and Presentation

Data will be presented in a table summarising the evidence. A narrative summary will accompany the results. Categories for data presentation include: year of publication, author(s), country, setting, population and sample size, research design and methods, intervention type, outcome measures, and impact and results. A lay summary will be co-produced with the PPI group.

Work package 2 – Feasibility study

Aim:

To explore the acceptability, feasibility and fidelity of an RCT exploring a hybrid format of group exercise to support sustained physical activity habits in people with Parkinson's Disease.

Objectives:

- identify ways to ensure representative recruitment and engagement
- monitor adherence to the intervention
- clarify the time needed to collect, clean and analyse data
- confirm the choice of secondary outcomes and their characteristics
- evaluate the practicality of delivering the intervention in the proposed setting
- explore the signal of efficacy of intervention

Methodology

Study design:

A feasibility study is chosen over a pilot study as it will ascertain that this novel intervention can be delivered as an RCT.[38] It will be a non-randomised, single-arm, feasibility study with pre and post-test design.

Sampling:

Convenience sampling will be used to recruit 25-50 participants.[39] To achieve a sample size at the upper end of target it is estimated 150 eligible participants will need to be screened accounting for a 35% participation rate to within a 95% confidence interval of +/- 10%.[40] This is based on a recent pilot study of PD participants who reported a 35% participation rate and 11% attrition rate.[41]

Inclusion criteria:

- Idiopathic PD
- Stable medication for >3/52
- Hoehn & Yahr (PD rating scale) I-III
- Mild to severe gait disturbance with a score of ≤2 on the Movement Disorders Society-Unified Parkinson's disease Rating Scale (MDS-UPDRS) item 29
- Aged 18+
- Safe to do physical exercise at home.

Exclusion criteria:

- Atypical Parkinsonism
- Cognitive impairment (MMSE score of <21/30)
- Unable to walk independently 10 m without aids or assistance.
- Severe, unpredictable episodes of motor fluctuation
- Comorbidities preventing safe exercise e.g., heart or lung conditions, uncontrolled diabetes, other neurological disorders, cancer or musculoskeletal conditions e.g., back pain or severe arthritis.

Recruitment will be carried out in partnership with NHS general practices, neurology clinics and therapy services in

a North London borough. They will identify eligible PwP who meet inclusion criteria and provide them with a participant information sheet. Interested participants will register their interest via telephone or email. Screening and eligibility checking will be conducted by the principal investigator before obtaining informed consent.

The hybrid model offers greater inclusivity for PwP to participate in the study by allowing flexibility in participants mode of attendance. To support recruitment of PwP from Black, Asian and Mixed Ethnicities (BAME), often under-represented in research, I will collaborate with general practices with higher BAME populations as recruitment sites and work with the local Parkinson's UK support group to identify a trusted advocate who can support accessing local communities to inform about the project. To support participants from lower socioeconomic groups, keen to participate but lack access to IT equipment or internet, tablets and dongles will be provided for the duration of their intervention.

Intervention:

I have developed this novel intervention through an iterative process combining theory, evidence-based clinical practice, and working with the PPI group to increase the likelihood of behaviour change and intervention implementation.[42,43] Intervention mapping, logic modelling, theories of implementation science and the COM-B model of behaviour change were used to develop the intervention.[44] Within PPI meetings, tasks encouraged members to consider important exercise components and self-management concepts to inform the intervention. The intervention will be refined with input from academic and clinical supervisors, PPI and findings from WP1.

The intervention consists of a 90-minute exercise and peer self-management group over 12 weeks designed to build exercise self-efficacy and support behaviour change. Participants have flexibility to attend in-person or virtually through Microsoft Teams. Each group will have a maximum of 12 participants, aligning with best practice guidelines regarding health & safety within the gym space and for online management. To ensure participant safety when exercising remotely a home risk assessment will be completed following guidance from Parkinson's UK and the CIMSPA policy for 'safe delivery of exercise at home'.[45,46] The home risk assessment will advise on suitable environments to exercise and technological equipment setup and guidance.

Weeks 1 & 12 consist of in-person completion of outcome measures. Week 1 will introduce the exercises and class structure. Participants will be provided a booklet containing tools to support them during the intervention including outcome measures, PA diary, exercise guides, and resources to facilitate exercise self-management e.g., education, reflective diaries, goal-setting and problem-solving worksheets. The exercise component will last 60 minutes (10-minute warm up, 40-minute circuit-based exercise and 10-minute cool down). Collaboration with the PPI group informed the exercise content which will be refined and finalised upon findings from WP1. Exercises will encompass strength, aerobic, balance, cognitive, gait, and functional components, in dual and single task activities as per guidelines.[5] Participants will be encouraged to work at a moderate-high intensity (60-80% rating of perceived effort). Recordings of the exercises will be emailed to participants to support independent exercise. Two instructors will lead the class, provide feedback, refine technique and progress participants in-person and online as appropriate.

Following the exercise component, a 30-minute peer self-management programme will be facilitated by the instructor. Peer learning has been shown to enhance self-efficacy through shared problem-solving, activity habits and successes which may support sustained PA habits.[47] Self-management programmes embedded in self-efficacy principles and devised with PPI have been shown to be more successful in their implementation.[48] Collaboration with the PPI group supported development of the self-management programme. Topics are embedded in behaviour change theory and based on established interventions including reflection, self-discovery, self-monitoring, problem-solving and activity and goal setting to support physical behaviours and cognitive process.[49,50] Content will be refined during year 1 of the DCAF through an iterative approach in collaboration with PPI and academic supervisors.

Data collection:

A mixed methods approach will measure the acceptability, feasibility and fidelity of a future RCT. This approach will enable triangulation, completeness and explanation when considering feasibility outcomes. Participant burden was considered when finalising the number of outcome measures.

Outcome measures and their relation to the study aims and objectives are attached in Table 1.

<u>Primary outcomes:</u>

Acceptability: Demographic data including age, socioeconomic status, education level and ethnicity of those screened will be collected to review representative engagement. Prospective acceptability using the Theoretical Framework of Acceptability (TFA) questionnaire will explore eligible participants' reasons for non-participation or

dropping out.[51,52] The TFA was developed to enable acceptability to be assessed in the design, evaluation, and implementation phases of healthcare interventions.[52] The questionnaire will contain both open and closed questions. All participants who complete the intervention and practitioners involved in delivering the intervention will receive the questionnaire and be invited for a semi-structured interview to explore acceptability and experiences of the study and intervention. Interviews will be conducted by the principal investigator utilising a topic guide using the TFA.[52] Interviews will be conducted at a location convenient to the participant and researcher, recorded via digital voice recorder and fully transcribed. A fieldwork journal will be kept when undertaking interviews to note additional relevant details which arise. This data will add context to findings as a form of triangulation. Transcripts will be sent to interviewees to confirm accuracy.

Feasibility: Will be assessed by collecting data on recruitment, adherence, completion rates and adverse events. Recruitment rate will be derived from eligibility and consent rate. Adherence will be derived from the number of attended sessions per participant. Completion rate will be derived from retention of participants from start to primary end point and the number of completed datasets. Adverse events during the intervention period will be reported in a log and provided to the Sponsor. Severe adverse events including those leading to hospitalisation (e.g., falls), life threatening events (e.g., cardiac events) and death will be reported immediately to the Sponsor using a reporting form and the study paused until considered safe to continue or terminated.

Fidelity: Will be assessed through interviews with participants and practitioners and through direct observation of six group exercise sessions by the principal investigator to establish whether the intervention is delivered as planned against a checklist of core intervention components.[53] Fieldnotes will be kept by the observer which will be used to add context and explanation to findings as a form of triangulation.

Secondary outcomes:

The PPI group explored outcomes considered important to measure before and after the intervention. This influenced the choice of secondary outcomes. Signal of efficacy will be assessed through effect size of secondary outcomes measured in-person at baseline (week 1) & post-intervention (week 12) at the exercise venue. This will provide preliminary data on PA, functional mobility and self-efficacy to inform sample size calculations for a future RCT.

Physical activity outcomes

Physical activity diaries: PA diaries are frequently used to record and measure levels of PA in health population research due to their practicality, low cost, low participant burden, and general acceptance.

Body-worn accelerometer: The GeneActiv™ monitor from Active Insights is a body worn triaxial accelerometer that monitors PA and activity intensity (Activinsights.com). Triaxial accelerometers have been shown to accurately measure PA in PwP and are tolerated well by participants.[54] The accelerometer will be worn during the period of the intervention.

Functional mobility outcomes

Timed Up & Go (TUG): One of the most used and reported measures for functional mobility in PwP. It measures the time needed to rise from an armchair, walk three meters, turn, walk back and sit down again. Longer times to complete the TUG are associated with increased fall risk. The TUG has demonstrated satisfactory clinimetric properties in PwP.[55] It has also shown ability to discriminate between early and middle disease stages. The test is easy to administer and is useful for evaluating intervention effects in ambulatory PwP.

Self-efficacy outcomes

Self-Efficacy for Exercise scale (SEE): Based on work by McAuley's 1990 (unpublished)[56] self-efficacy barriers to exercise measure. It's a 9-item self-rating instrument measuring self-efficacy through belief in the ability to continue exercising in the face of barriers in sedentary adults in the community. The ratings are graded on a percentage scale from 0-100%. There is statistically significant correlation between efficacy expectations related to exercise and adherence to a regular exercise program ($r = .42$; $p < 0.05$).[57] There is evidence for the reliability and validity of the SEE scale for use with diverse populations.[58]

Importance/Confidence measure: To support health outcomes, measures of patients' readiness to change is recommended in clinical practice.[59] One aspect of readiness is the confidence in the individual's ability to succeed and importance placed on changing.[59] Both are linked to self-efficacy theory. Single measure visual analogue scales can be used to measure both importance and confidence to self-manage PA.[59]

Data analysis:

Interview data will be analysed using NVivo software[60] following a framework analysis approach to aid triangulation of findings with the TFA questionnaire and field notes. An inductive and deductive approach will lead to development of a coding framework based on the 8 TFA constructs. Coding and categorisation of data will be reviewed by a second researcher. Themes derived from data analysis will be sent to participants for corroboration and any disagreements will be resolved through discussion.

Demographic data will be presented descriptively including means (with standard deviation) for any continuous data and raw count (percentages) for categorical data. Primary and secondary outcomes will be analysed using simple descriptive statistics. Findings will be presented as medians (range) for ordinal data, mean (95% confidence interval (CI)) for continuous data and raw count (number, %) for nominal data. A Cohen d test will measure effect size between secondary outcomes at T1 (baseline) and T2 (12 weeks) timepoints to guide future sample size calculation. All data analysis will be completed using SPSS.[61]

Quantitative and qualitative findings will enable conclusions and inferences about the feasibility of a future RCT to be drawn. Refinement of the proposed study will support optimal conduct for a future RCT.[62] Triangulation will be used to analyse acceptability of the study through questionnaire acceptability ratings and qualitative data from the questionnaire open-ended questions and semi-structured interviews. Integrative analysis using a comparison joint display regarding acceptability will be employed.[63] To analyse recruitment, data on recruitment rate and individual rationale for study participation from interviews will provide completeness of data. Adherence will be analysed through completion rate, attrition rate, field notes and open-ended questionnaire responses from participants who drop out of the study. Integrative analysis using a synthesis joint display with the complementary quantitative and qualitative data findings will address this specific domain.[63] Fidelity outcomes will include quantitative data on the fidelity measure checklist alongside field notes and participants' and professionals' interview data. This will provide explanation for any differences in participant or professionals' characteristics and context on barriers or facilitators to delivering the study. Integrative analysis using an interconnection joint display will uncover differential patterns within this domain.[70]

Progression criteria:

A traffic light system will establish potential progression to an RCT (≥80% proceed, ≥60 - ≤79% proceed with rescue package, ≤59% not proceed and return to development and refining the intervention design). The progression criteria to be used are the following: (a) participant recruitment, (b) mean participant adherence, (c) completed data sets, (d) participant and instructor acceptability and (e)consideration of adverse events.

Dissemination, Outputs and anticipated Impact

What do you intend to produce from your research?

I will register and publish study protocols for WP1 & 2. Following completion of both WPs I will produce peer-reviewed publications. To reach wider audiences I will present the findings of both WPs at relevant conferences. If WP2 findings meet progression criteria I will produce a protocol for a future pilot study/RCT.

How will you inform and engage service users, services and the wider population about your work?

World Parkinson's Congress (WPC) brings together academics, clinicians and PwP as a community. I will present findings from my research at the WPC 2026 conference. To inform PwP, who have less access to scientific papers and conferences, I will feedback through webinars with Parkinson's UK charity, presentation at a Parkinson's UK support group within the recruitment area during years 2 & 3 of my DCAF. I will work with my advisory groups during times when outputs and dissemination are occurring to ensure content is appropriate to PwP and the wider population.

How will your outputs enter our health and social care system or society as a whole?

I will produce accessible infographic posters as well as guidance leaflets for healthcare professionals and PwP on hybrid forms of group exercise. These will be disseminated through my professional social media (Twitter), academic platform (ResearchGate) and through the Parkinson's UK exercise Network. I will present each WP findings at a UCL Partners Centre for Neurorehabilitation seminar which provide an opportunity for researchers within UCL partners to exchange ideas and insights. Through collaborations with the Parkinson's Excellence Network and Professor Bas Bloem (founder of ParkinsonNet who developed the European Physiotherapy guidelines for Parkinson's disease) I hope to influence policy and guidance on exercise provision for PwP.

What further funding or support will be required if this research is successful?

I will seek further NIHR (ACAF or HTA) or charitable funding for post-doctoral study to complete a systematic review to further explore online exercise groups, and a pilot study or RCT of the hybrid exercise group intervention.

What are the possible barriers for further research, development, adoption and implementation?

A lack of research funding and/or clinical academic career positions to enable continued clinical research and development of a clinical academic career. Another possible barrier includes any changes to current licenses for NHS staff to use Microsoft Teams software. Lack of support including commissioning of hybrid exercise groups could be a barrier to uptake of findings into the health and care services.

What do you think the impact of your research will be and for whom?

Whilst there will be no immediate benefit from this project to patients and healthcare professionals the findings will inform a future RCT which will lead to patient benefits. This will be through demonstration of increased self-efficacy to sustain physical activity habits which has previously been shown to improve health-related outcomes for people with PD.

Following a future RCT the hybrid intervention could be applied or adapted to outpatient and community services who support PwP and other long-term conditions as well as private healthcare and exercise providers. PwP in England have higher rates of A&E admissions, prolonged length of hospital stays and in-hospital mortality.[13] Much of this related to physical inactivity. Findings from this research could increase access to and support sustained adherence to PA for PwP and other neurological conditions. PD is a good model for developing community-based exercise and self-management programmes suitable for other neurological/degenerative conditions.[29]

Project Management

I will undertake managing the DCAF as principal investigator with academic and clinical support from my supervisors building on the strength and success of our team during my PCAF. Monthly meetings with my primary supervisor and bimonthly meetings with my both academic supervisors will provide ongoing guidance during the DCAF. At least quarterly meetings with the PPI and advisory groups will provide guidance and hold accountability to manage the project to time.

The project will be managed through efficient time planning and management ensured through adherence to my DCAF GANTT. An element of slack has been included in timings to mitigate for potential delays. I will use a weekly journal to specify activities to be completed and break down larger chunks of work into smaller work packages. Completed tasks will be checked off and actions stated. This will be reviewed at each monthly supervisor meeting. I will use Microsoft Project software for this.

The financial management of the award will be managed through the research and departmental administrators of the Department of Clinical and Movement Neurosciences, UCL.

Ethics

WP1: Ethical approval not required. No ethical concerns identified.

WP2: Ethical approval will be sought prior to commencement of the study through the Health Research Authority through the Integrated Research Application System. Participants will provide signed informed consent. Participation or not will not affect input from NHS care services. Data will be stored in a password protected computer and comply with GDPR guidelines. A Risk assessment will consider the study risks. Participants will have a process to raise comments and complaints that may arise.

Success Criteria

Measurements of success

- Registration and publication of protocols for WP1 & 2
- Completion of both work packages
- Publication of WP1 & 2
- Completion of training and development programme
- Successful completion of dissertation and defence of PhD

Key risks to delivering this research.

Risks	Contingencies
Unable to recruit sufficient participants and staff to WP2	• Work with North Thames CRN. • Utilise existing knowledge and contacts from recruitment area. • Utilise support from supervisors who are clinicians working in NHS sites in the recruitment area. • Costing for incentives to participation and cost of attendance.
Training & Development courses not running as planned	• I will work closely with academic and clinical experts to identify suitable alternatives should planned training be cancelled. This could include academic training provided by UCL. I will also seek relevant academic support from within my supervisory structure.
Delay in IRAS ethical approval for WP2	• Utilise SoECAT and feasibility study protocol to support IRAS application. • Prompt ethical application to IRAS to provide sufficient time for any amendments.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

During my PCAF I collaborated with Parkinson's UK and established an advisory group for the development of the DCAF application and research proposals. The advisory group consists of 4 people with Parkinson's disease and one family member/carer from all over the UK. They were considered appropriate to involve due to their relevant and direct experience of PD. It's based on mutual respect and shared values on the role of exercise for PD. There have been 3 online meetings focusing on different aspects of the research proposal, work packages and DCAF application. To support effective PPI I undertook training from the NIHR of how to conduct PPI: focus groups and more.

Meeting 1: Focused on getting to know each other and exploring the topic idea, research question, aims and objectives. I was able to recognise the interest and value of the research topic as all participants felt it was timely and important research. Discussions focused my attention to consider equal opportunity and how to measure accessibility to the feasibility study, including participants from lower socioeconomic backgrounds or those with low exercise self-efficacy. From this I refined the recruitment process for the WP1.

Meeting 2: Contributed to development of the logic modelling by identifying what participants felt was important to include in the feasibility study exercise intervention. This enabled me to consider exercise type, class structure and equipment required to include in costing of the study. We also discussed what outcome measures were important and meaningful to assess. This enabled me to carefully select secondary outcome measures that matched with these discussions.

Meeting 3: This focused on feeding back to participants on the DCAF aims and WPs 1 & 2 to collaboratively create the project title. Discussions on barriers to self-managed physical activity contributed to the development of the self-management programme content in WP2.

The advisory group have also been involved in the writing of the plain English summary. Separately to the advisory group I have presented the DCAF proposal at the Parkinson's UK South East Research Interest Group Autumn meeting. Feedback from this was positive towards the study aims and objectives.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

During my DCAF I will continue working with the PPI group, expand the group to more members through support from Parkinson's UK. The aim is to include PwP who are keen to exercise but struggle to do so or are less confident with online formats of exercise. This will help to introduce insights and perspectives to refine WP2 acceptability to PwP and recruitment strategies. Due to the length of the project and unpredictability of PD, membership of the PPI group will be rolling to allow flexibility to participate. The advisory group will continue to meet 3x yearly online. Meetings will have pre-set agendas which will be forwarded to participants prior. Meetings will be recorded so information can be reviewed and revisited at later time points. All people involved will be appropriately recompensed for their time and provided with training they need, in accordance with NIHR PPI guidance.

The aim of early PPI input is to consider recruitment strategies for WP2 including language use in advertising and participant information sheets. It will consider how to identify and encourage recruitment of PwP from lower socioeconomic backgrounds and those with low exercise self-efficacy. PPI input will also be to continue to refine the intervention content (exercise and self-management programme) of WP2. This will be considered alongside findings from WP1.

Additionally, PPI input will be used to support study documentation. This includes reviewing the language and content of the participant booklet for WP2. There will also be input to aid completion of the ethics application for WP2 including writing of the plain English summary and participant information sheet.

Finally, PPI will help with dissemination of the findings from WP1 and WP2 through review of summary documents and infographics prior to being sent to participants or for public dissemination.

I have established an advisory group of academics and healthcare professionals. Although this is not the same as PPI, this will allow for perspectives from clinical academics and healthcare professionals which may assist with supporting acceptability of the feasibility study to healthcare professionals.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Not applicable

8. Training & Development and Research Support

Proposed training and development programme

Academic training and development:

I have utilised the Vitae research development framework (vRDF) to identify skills and capabilities for the development of a clinical academic and my proposed research. My training and development plan builds on my existing skills and knowledge to support me to conduct my research plan.

UCL has a proven track record supporting doctoral researchers with postgraduate training. The UCL Doctoral Skills Development Programme (DSDP) provides free training on topics embedded within the vRDF to develop skills for research, professional development, and employment. I will strategically organise my time over the 3 years to best support my research needs and leadership development and attend the following training (not exhaustive):

- **Knowledge & intellectual abilities**: Creativity in research; Introduction to EndNote; Reading and literature review.
- **Personal effectiveness**: Influencing & negotiating; Managing as a researcher; Producing thesis and reports.
- **Research governance & organisation**: Transparency in research, Grant and charity funding.
- **Engagement, influence & impact**: Writing style and process; Planning for impact.

The UCL Queen Square Institute of Neurology (IoN) is home to some of the world's most influential academics across disciplines in clinical and movement neurosciences, clinical neurology and neuropathology. It supports academic development through research groups, a frequent varied seminar series and informal peer-support and several support systems in place for PhD students including an induction programme and buddy scheme. Its strong clinical partnership with the National Hospital for Neurology and Neurosurgery and Royal Free Hospital gives me access to high quality neuroscience teaching through integration of research, clinical practice and education.

Academic courses relevant to research project

- Joanna Briggs Institute (JBI) scoping review 1-day in-person workshop. To enable exploration of the theories and concepts relating to scoping reviews and other types of evidence synthesis and equip myself with the knowledge and tools required to successfully undertake and report WP1.
- Pilot and Feasibility Studies 1-day in-person workshop (Queen Mary, UoL). To enable me to have greater understanding in the planning and reporting of pilot or feasibility studies. This will support me to plan, undertake and report on WP2.
- Randomised controlled trial course, 5 days in-person (Oxford University). To provide thorough grounding in the principles and practice of RCTs for evaluating healthcare interventions. This will support development of a protocol for post-doctoral research from completion of WP2.
- Implementation Science 2-day in-person Masterclass (KCL). To facilitate knowledge of implementation science and related interdisciplinary perspectives to support protocol development of the complex intervention (WP2).
- Emerging Clinical Leaders. 6 days virtual attendance over 6 months (King's Fund). To develop skills to succeed in the modern health and care environment, collaborating with colleagues from other disciplines and build my career as a clinical academic.

Personal and professional practitioner training and development:

I will undertake the following clinical training and attend these relevant conferences to support my development as a world-leading clinician in the field of PD. The courses will allow me to teach my clinical colleagues best practice for this client group also.

- Parkinson's Disease and Movement Disorder Course, University of Salford. A 30 credit blended distant learning module over 1 semester to enable myself to systematically and critically apply theory and practice in the management and care of people with movement disorders acting with clinical autonomy. It requires 3 days on-campus attendance.
- PWR!Moves Therapist Certification 2-day in-person Workshop, Tuscon, Arizona. To master this emerging evidence-based therapeutic concept which utilises progressive exercise and the PWR!Moves for PD-specific functional skill training across a range of disease severities and real-world environments.
- Atypical Parkinson's 1 day in-person course (NCORE). To gain an increased knowledge and

understanding of these conditions including therapy management and person-centred care.

- Parkinson's: Train the trainer Course (Parkinson's UK, 3hrs virtual attendance). A course for experienced health professionals to become a Parkinson's Excellence Network trained trainer & learn how to deliver PD training to their colleagues.

Conferences

Year 1: Movement Disorders Society (MDS) Conference 2024. Philadelphia, USA. An opportunity to meet and network with PD research and clinical specialists and including those in collaboration section of DCAF application.

Year 2: World Congress of Physiotherapy Therapy (WCPT) 2025. Tokyo, Japan. An opportunity to network with physiotherapy clinical specialists, including those interested in neurology and PD and present findings from WP1.

Year 3: World Parkinson's congress (WPC) 2026. TBC. An opportunity to network with clinical specialists in PD and present preliminary findings from WP2.

Clinical practice to be undertaken

Further to my training, I will work in varied clinical practice roles to maximise my understanding of healthcare management for PwP throughout the care pathway. This will include working with NHS colleagues in neurology outpatient clinics and community services and with private practice colleagues supporting PwP through online exercise groups.

Year 1: Camden Neurology & Stroke Service (CNSS) (0.2fte). My current NHS community-based neurological rehabilitation team where I will continue to run community-based PD exercise groups. Working with CNSS will allow me to network and develop collaborations to recruit to WP2.

Year 2: Neuro Heroes (0.2fte). A private online company providing 1:1 and group-based exercise for people with PD and other neurological conditions. Working with Neuro Heroes will develop my organisational, management and clinical skills in delivering online exercise.

Year 3: Royal Free Hospital (RFH) & National Hospital for Neurology & Neurosurgery (NHNN) (0.2fte). Working in two busy neurological outpatient departments run by specialist neurologists, nurse specialists and physiotherapists will develop clinical skills to support PwP recently diagnosed and at different stages of their disease compared to the CNSS. Working with CNSS will also allow me to network and develop collaborations to recruit to WP2.

Timetable

	Conferences	Academic training		Clinical training	Clinical practice
Year 1	MDS conference, Sept 2024	UCL DSDP training	JBI scoping review workshop Queen Mary UoL, pilot & feasibility studies workshop	PD and Movement Disorders Course, Uni of Salford PWR!Moves workshop, Tuscon, Arizona	CNSS
Year 2	WCPT, May 2025		Oxford Uni, RCT course KCL implementation science masterclass	Atypical Parkinson's Course, NCORE Parkinson's: Train the trainer Course, Parkinson's UK	Neuro Heroes
Year 3	WPC, 2026 (TBC)		King's Fund, Emerging clinical leaders course		RFH & NHNN

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Professor Anette Schrag

Institution	University College London

Position	Professor of Clinical Neurosciences

Description of support provided
Professor Schrag is a professor of clinical neurosciences within the department of clinical and movement neurosciences at the UCL/Queen square Institute of Neurology. Professor Schrag's research focuses on PD and other movement disorders. Her current research has focused on patient-centred healthcare, with work concentrating on patient-reported outcomes and quality of life. She received the Silver Medal of the European Society for Person Centred Healthcare in 2018 related to this work. She is currently running several clinical trials as Chief Investigator in the NHS including PD-Care (Development of a facilitated Self-Management Tool for patients with Parkinson's). Prof Schrag has supervised nine PhD students, five of whom as primary supervisor (three completed, two ongoing), and four as secondary supervisor (two completed, two ongoing), one medical doctorate student (ongoing) and 21 Master students (20 completed, one ongoing). Prof Schrag and I share an interest in exercise and self-management interventions for PD. We have already established a constructive working relationship during my PCAF and she is keen to continue as my primary supervisor during my DCAF. We have maintained virtual meetings every 2 months and plan to meet monthly during my doctoral studies. She has experience of a wide variety of research methodologies including scoping review and clinical trial which will be valuable for supporting my research plan. Prof Schrag has an extensive network of peers and colleagues nationally and internationally, including Prof Bas Bloem, which will be of substantial benefit for developing collaborations with world-renowned clinical and academic experts.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Professor Fiona Jones

Institution	St George's University, London

Position	Professor of Rehabilitation Research

Description of support provided
Professor Fiona Jones is Professor of Rehabilitation Research at St George's University of London and the founder and CEO of 'Bridges self-management' which works alongside people with long-term conditions to design and deliver best practice in self-management. Her research interests lie in neurorehabilitation and self-management specifically inclusive approaches to self-management support and sustainability within rehabilitation settings. She is leading and collaborating on several research projects related to personalised self-management support including the NIHR funded Listen project (Long COVID Personalised Self-management support- co-design and Evaluation). Fiona has supervised 8 PhD students to completion. She has also experience in PhD examination; upgrade MPhil to PhD (3 students) and PhD viva (13 students). We share interests in clinical neurological research and although her work focuses predominantly on stroke it is easily transferrable across different neurological populations. Prof Jones's has supported me, alongside Prof Schrag, as my secondary academic supervisor during my PCAF. We have met as a research team every 2

months and Prof Jones has provided valuable support for the development of the two work packages, especially with guidance on logic modelling to develop the feasibility study. Meetings will continue during my DCAF. Prof Jones' expertise in self-management and her experience of designing interventions for addressing physical inactivity will undoubtedly be of value during my doctoral studies. Prof Jones will also provide support to help develop my route to achieving high research impact and skills to communicate and disseminate my ideas and research findings to different stakeholders.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Dr Caroline Appel

Institution	Central & North West London NHS Foundation Trust

Position	Lead Physiotherapist Camden Neurology & Stroke Service

Description of support provided
Dr Caroline Appel is the clinical lead physiotherapist within the Camden Neurology & Stroke Service (CNSS), an NHS community neurological rehabilitation team. Caroline has been working as a clinical academic since she completed her PhD in 2015. Her thesis explored shoulder strapping for stroke-related upper limb impairment. Caroline understands the difficulties in developing a clinical academic career in the NHS and barriers to conducting clinical research in the NHS. Caroline and I have worked together for over 5 years, and we have an excellent professional relationship. We have been working together to build research capacity into the CNSS which Caroline has taken a lead on. Caroline has been supportive of my clinical academic career and supported me in successfully applying for grants. She has also regularly attended PCAF supervisory meetings to contribute clinical advice on my research plan. Caroline has always supported my clinical, managerial and professional opportunities. Caroline has collaborated with the CSP to promote the 'Making Activity Happen' and 'Love Activity, Hate Exercise' campaigns. She also added to the CSP's objective to advocate for the profession to key stakeholders and the general public through contribution to a BBC article on regaining strength after Lockdown. Caroline is a trustee of LEGS charity and a founding member of Long COVID Physio and which aims to provide peer support for healthcare workers living with Long COVID. Her breadth of roles and experiences in public sector, charitable and regulatory bodies will provide broad perspectives for my career development as a clinical academic.

Collaborations
Professor Bastian Bloem, Radboud UMC, Nijmegen, Netherlands. Professor Bloem is a world-leading expert on Parkinson's disease (PD). He founded ParkinsonNet, the world's foremost healthcare model for people with PD (PwP). His research interest lies in identifying new ways of slowing the progression of PD and ensuring optimal quality of life for patients and their families. Physiotherapy and exercise are important in his approach and he has advocated the role of online exercise for PwP. Professor Bloem sees creating partnerships as the ideal way to continue to innovating and improving care for PwP and he is keen to foster a collaboration with myself and support my studies. His expertise will also help development of the future RCT, consideration of measuring effectiveness through laboratory-based outcomes as well as someone to establish multi-centre site collaborations. **Dr Rebecca Farley**, Parkinson's Wellness Recovery PWR!, Tucson, Arizona. Dr Farley is a physiotherapist, neuroscientist, and PD exercise specialist. She is the Chief Scientific Officer and Founder of Parkinson Wellness Recovery PWR!, PD-specific in-person and online functional group exercise programmes. Her current focus is the publication of research from her facility and integrating new research into workshops and her programmes. A visit

to PWR! will benefit me by attending the PWR!Moves workshop as specified in my training & development plan. A visit will enable me to translate approaches used by this private sector/industry to future clinical interventions and research within the NHS. Dr Farley and I have corresponded and she is keen to facilitate a visit.

Melissa McConaghy, Advance Rehab Centre, Sydney. Melissa is a Fellow of the Australian College of Physiotherapists and Specialist Neurological Physiotherapist. Melissa is an educator, clinician and author and strong believer in exercise for managing PD. She founded and curated the online global Parkinson's Conference, "INSIGHT into PD", and authored the book "The New Parkinson's Treatment: Exercise is Medicine". She founded PD Warrior, an exercise intervention approach for treating Parkinson's disease. Melissa and I have communicated and she welcomes a visit to build my clinical skills in teaching PD warrior exercises and discuss skills in engagement, influence and impact in disseminating my research and promoting myself as a clinical academic.

Neuro Heroes. Neuro Heroes is the UK's first online neuro gym, delivering condition specific exercise classes for PwP and other neurological conditions. I will work with them in a supernumerary position in year 2 of my DCAF as part of my clinical practice to be undertaken. This collaboration will facilitate shared knowledge of running online exercise groups for PwP and enable me to build working relationships between the NHS, private sector industry and academia.

Dr Julie Jones, Robert Gordon University (RGU), Aberdeen. Julie is a clinical academic fellow and a senior lecturer at RGU. She is a member of the Strategy Advisory board of Parkinson's UK in Scotland, chairs the Scotland wide Parkinson's Active project and is the co-lead of the Parkinson's Excellence Network Exercise hub. Julie completed her PhD exploring the feasibility of a collaborative approach to exercise delivery between the NHS and community sector. We have collaborated on previous research and plan to continue doing so. Julie has agreed to be involved in the planned clinical academic advisory group.

Dr Bhanu Ramaswamy. Bhanu is an independent Physiotherapy Consultant and Honorary Visiting Fellow Sheffield Hallam University. Bhanu is the lead for the service user involvement committee for the Parkinson's Excellence Network. We have met regularly during my PCAF and Bhanu has guided my DCAF application in an advisory role. Bhanu has agreed to be involved in the planned clinical academic advisory group.

Host Organisation Support Statement

The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

James started working in the Camden Neurology and Stroke Service (CNSS) within Central and North West London NHS Trust in 2018. He has been a strong clinician bringing extensive experience of working in diverse clinical neurological settings as well as academic skills to the team. James has lead on service development projects including the development of an exercise group for Parkinson's disease (PD). James will have a supernumerary clinical role during the first year of his DCAF working 0.2fte. This will allow him to continue developing clinical and leadership skills through delivering exercise groups for and supporting people with PD (PwP). He will also provide clinical skills training, share expert knowledge and guidance for managing PD within the Trust.

James has demonstrated a clear career path with interest in continuing clinical work whilst also promoting and supporting research. As the Clinical Lead Physiotherapist with a PhD myself, I support James to take every opportunity to build on his clinical academic skills during the DCAF. CNWL trust is keen to develop research from Allied Health Professions and has close links with University College London (UCL). It is committed to equality of opportunity and believes that through its EDI mission statement it can achieve high quality evidence-based research and clinical care. James has met with Dr Ryan Kemp (Director of therapies), Johanne Watson (Chief Physiotherapist) and Dr Cherry Kilbride (AHP research lead) to agree to work together and develop clinical academic career opportunities within CNWL for those who have undertaken academic training. The clinical and research skills James will acquire will be of immense value to our team and CNWL. We are keen to increase our research output as well as our influence within North Central London Integrated Care System.

UCL boasts itself as an outstanding place to study and support research. It is ranked as one of the top 10 universities in the world (QS world rankings 2023). UCL was rated 2nd in the UK for research strength in the 2021 Research Excellence Framework. UCL's research environment achieved the maximum possible score, 100% at 4*. Specific to James' area of interest, UCL is the 3rd most highly cited university in Europe for neuroscience and

behaviour studies (InCites scientific indicators 2018) and UCL biomedical and health researchers are the most influential in England, including the field of clinical neurology (RAND bibliometric analysis 2015). Moreover, our education, training and research is delivered in partnership with leading specialist hospitals with designated status as an Academic Health Sciences Centre. UCL prides itself on creating an inclusive culture for students and employees and received a silver Athena SWAN award in 2021.

Through UCL James will have access to world class resources, including a comprehensive doctoral training programme, 15 libraries and an excellent support network for research and academic writing. Doctoral students are supported via Thesis Committees throughout their PhD and report their progress on the Research Log. This method provides the student with wider intellectual contribution, access to copious expertise and an overview of the student's development at specific points during the degree. ‘UCL Researcher Initiatives’ supports early career researchers through a variety of learning, networking, and community listening opportunities.

James will be embedded academically in the UCL/Queen’s Square Institute of Neurology (IoN), Department of Clinical and Movement Neurosciences. The department is a world-leading group bringing together expertise across the clinical and basic science spectrum of the physiology and pathology of human movement and movement disorders. The IoN has strong links with clinical service delivery working collaboratively with several NHS providers to promote the translation of research that is of direct clinical relevance into improved patient care and treatment. Boasting clinical and applied scientific activity the Institute is a unique national resource for postgraduate training and research in neurology, including James’ interest of PD. Our mission is to provide the best quality learning and working experience, nurturing staff and students through being an inclusive and supportive workplace to reach their full potential. The institute supports clinical academic research careers through offering a wide range of training courses by the Graduate School and local department, as well as encouraging attendance at conferences and external meetings. The Institute offers a weekly teaching programme which James will be able to access as needed as a doctoral student. Students are given the opportunity to present their research, alongside their peers studying for PhDs, at the well-regarded annual Queen Square Symposium.

James’ chosen area of research concerns the feasibility of a hybrid exercise programme for people with PD. James will be supervised by Prof Anette Schrag who has extensive research experience in PD and clinical trials and alongside Professor Fiona Jones they will be able to support completion of his work packages and training & development plan. Students at the IoN are ideally placed to exploit the opportunities for career progression which arise from studying in a well-funded national and international centre of excellence. At UCL we are proud to be one of few institutions that has supported and funded clinical academic partnerships. Our intention is to develop the necessary skills, experience and attitudes for those who wish to pursue careers as clinical academics to do so. It is with great excitement that we can say we will support James to undertake a DCAF and supported by CNWL trust, foster a long-term partnership whereby a highly regarded academic institution is supporting clinically relevant research.

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	James Alexander

Employing organisation	University College London

Grade	CL8

Starting point on scale	08

Increment Date	01/08/2024

Justification of Costs
Previously employed at top band 7 with inner London weighting on agenda for change scale within NHS. This salary has been mapped across to equivalent salary on UCL pay bands. The matches to UCL clinical academic grade CL7 point 8.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	52430
Geographical Weighting (£)	4200
Other allowances (£)	0
Superannuation (£)	12515
Nat. Insurance (£)	6559
Current Annual Costs (£)	**75704**

2

Staff type	Shared Staff

Name	Research Assistant - Rehab assistant

Employing organisation	University College London

Role on this research	Rehab assistant

Grade	6B

Starting point on scale	25

Increment Date	01/08/2024

Justification of Costs
Rehabilitation assistant for running of intervention during WP2. Costed as NHS agenda for change band 4 at 0.1wte for 18 months

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	32108
Geographical Weighting (£)	4200
Other allowances (£)	
Superannuation (£)	8024
Nat. Insurance (£)	3755
Current Annual Costs (£)	**48087**

3

Staff type	Shared Staff

Name	Senior Research Fellow - Physio

Employing organisation	University College London

Role on this research	Physio

Grade	8

Starting point on scale	39

Increment Date	01/08/2024

Justification of Costs
Recruitment of physiotherapist for running of intervention during WP2. This has been costed at NHS agenda for change band 7 salary for 0.1wte for 18 months. Due to UCL costing the salary has been matched to equivalent UCL salary - non-clinical grade 8 salary point 39.

Details of Posts and Salaries Costs

	Total (£)

	Total (£)
Salary (£)	47964
Geographical Weighting (£)	4200
Other allowances (£)	
Superannuation (£)	11528
Nat. Insurance (£)	5943
Current Annual Costs (£)	**69635**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Mr James Alexander
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: Previously employed as a band 7 physiotherapist within the NHS. As of April 2024, beginning of fellowship, i would be on top pay step band 7. The equivalent salary has been mapped across to UCL non-consultant clinical academic pay scale and includes normal increments on that scale

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	52430	56845	61849	171124
National Insurance	6559	7168	7859	21586
Superannuation	12515	13491	14597	40603
Geographical weighting / Other allowances (if applicable)	4200	4200	4200	12600

Staff Type: Other
Other Type: Shared Staff
Name: **Research assistant - rehabilitation assistant for WP2**
Type of cost: HEI
% Full-time on this research: 10.0
Total months on this research: 18.0
Justification of Cost: Agenda for change band 4 rehabilitation assistant to support running of intervention during WP2 (0.1wte for 18 months in total). Costing has been matched over to equivalent UCL pay scale for purpose of costing the project.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	4809	2446	0	7255

Staff Type: Other
Other Type: Shared Staff
Name: **Senior research fellow - physiotherapist for WP2**
Type of cost: HEI
% Full-time on this research: 10.0
Total months on this research: 18.0
Justification of Cost: -

	Year 1	Year 2	Year 3	Total Cost (£)

Basic Salary	6964	3545	0	10509

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	196730	0	14211	210941
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Fellow travel costs international
Justification of Cost: All travel to be booked through UCL approved provider and in line with UCL travel policy. International travel includes: Year 1. Return flights to Tuscon, Arizona USA for collaboration visit with Dr Becky Farley and to attend PWR!Moves training. In line with UCL Expenses policy, a flight greater than 5 hours in duration qualifies for "premium economy" costs. Based on current internet search this is costed at £1,500. Year 2. Return flights to Sydney, Australia for collaboration visit with Melissa McConaghy. In line with UCL Expenses policy, a flight greater than 5 hours in duration qualifies for "premium economy" costs. Based on current internet search this is costed at £2,000. Year 3. Economy return train travel from London to Nijmegen for collaboration visit with Professor Bloem costed at £200.

	Year 1	Year 2	Year 3	Total Cost (£)
	1500	2000	200	3700

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Fellow travel costs UK
Justification of Cost: To be booked through UCL approved provider and in line with UCL travel policy. Economy rail. Includes: Year 1: Feasibility study workshop Oxford University, 1 x return rail journey = £80. University of Salford Parkinson's course 3x return rail journey 3x £133 = £399. Year 2: RCT course Oxford University. 1x return rail journey = £80. Atypical Parkinson's course, Derby. 1x Return rail journey = £80. Total = £639

	Year 1	Year 2	Year 3	Total Cost (£)
	479	160	0	639

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Intervention participant travel costs
Justification of Cost: Participant travel costs for travel to intervention venue for WP2 based on Transport for London zone 1-3 off-peak travel charge = £5.40 return (£5.40 x 12 weeks)x50 participants = £3,240

	Year 1	Year 2	Year 3	Total Cost (£)
	0	2592	648	3240

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference fees
Justification of Cost: Attendance at national and international conferences is essential to build my network and disseminate my research for maximum impact. I have specifically chosen related conferences to attend each year and have spread the costs evenly over each conference and year up to the guidance budget limit. Conferences planned to attend include: Year 1: Movement Disorders Society (MDS) Conference 2024 – Philadelphia, USA. Year 2: World Congress of Physiotherapy Therapy (WCPT) 2025 – Tokyo, Japan. Year 3: World Parkinson's

congress (WPC) 2026 – TBC. An opportunity to network with clinical specialists in PD and present preliminary findings from WP2.

	Year 1	Year 2	Year 3	Total Cost (£)
	200	200	200	600

Type of cost: HEI
Description: Journey Costs
Conference: True
Item Description: Conference travel costs
Justification of Cost: Attendance at national and international conferences is essential to build my network and disseminate my research for maximum impact. I have specifically chosen related conferences to attend each year and have spread the costs evenly over each conference and year up to the guidance budget limit. Conferences planned to attend include: Year 1: Movement Disorders Society (MDS) Conference 2024 – Philadelphia, USA. Year 2: World Congress of Physiotherapy Therapy (WCPT) 2025 – Tokyo, Japan. Year 3: World Parkinson's congress (WPC) 2026 – TBC. An opportunity to network with clinical specialists in PD and present preliminary findings from WP2.

	Year 1	Year 2	Year 3	Total Cost (£)
	500	500	500	1500

Type of cost: HEI
Description: Subsistence
Conference: True
Item Description: Conference accommodation and subsistence costs
Justification of Cost: Attendance at national and international conferences is essential to build my network and disseminate my research for maximum impact. I have specifically chosen related conferences to attend each year and have spread the costs evenly over each conference and year up to the guidance budget limit. Conferences planned to attend include: Year 1: Movement Disorders Society (MDS) Conference 2024 – Philadelphia, USA. Year 2: World Congress of Physiotherapy Therapy (WCPT) 2025 – Tokyo, Japan. Year 3: World Parkinson's congress (WPC) 2026 – TBC. An opportunity to network with clinical specialists in PD and present preliminary findings from WP2.

	Year 1	Year 2	Year 3	Total Cost (£)
	300	300	300	900

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Fellow London travel costs
Justification of Cost: Travel costs to central London (Zone 1-6 annual travelcard) for clinical practice roles during DCAF, all related travel within London for WP1 and WP2 (including observation visits, data collection, interviews) and all travel for training and development within London (KCL implementation course). Based on current Transport for London fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	2976	2976	2976	8928

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Fellow accommodation and subsistence UK
Justification of Cost: Accommodation and subsistence for international visits in line with UCL expenses policy. Calculation based on recent internet searches for nearby accommodation to training/site and subsistence calculated at £37/day. Year 1: Queen Mary feasibility workshop, Oxford, accommodation 1x (£150/night) plus subsistence 1x (£37/day) = £187. Uni. Salford Parkinson's course, Salford accommodation 3x (£120/night) plus subsistence 3x (£37/day) = £471. Year total = £658. Year 2: Oxford Uni RCT course, Oxford, accommodation 5x (£120/night) plus subsistence 5x (£37/day) = £785. KCL implementation science course (no accommodation or subsistence required as London based). Atypical Parkinson's course, Derby. Accommodation 1x (£120/night) plus subsistence 1x (£37/day) = £157. Year total = £942.

	Year 1	Year 2	Year 3	Total Cost (£)
	658	942	0	1600

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Fellow accommodation and subsistence international
Justification of Cost: Accommodation and subsistence for international visits in line with UCL expenses policy. Calculation based on recent internet searches for nearby accommodation to training/site and subsistence calculated at £37/day. Year 1 visit to Dr Rebecca Farley at PWR! Parkinson Wellness Recovery, Arizona, USA as part of training and development and collaborations. 5x nights (£150/night) and subsistence (£37/day) = £935. Year 2 visit to Melissa McConaghy at Advance Rehab Centre, Sydney, Australia as part of collaboration visit. 5x nights (£150/night) and subsistence (£37/day) =£935. Year 3 visit to Prof Bloem at Radboud UMC, Nijmegen, Netherlands as part of collaboration visit. 3x nights (£150/night) and subsistence (£37/day) = £561. Total

	Year 1	Year 2	Year 3	Total Cost (£)
	935	935	561	2431

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: PPI travel costs
Justification of Cost: Travel expenses for 3x in-person workshops (£50p/person x6 participants = £900 total (£300 each year of PhD)).

	Year 1	Year 2	Year 3	Total Cost (£)
	300	300	300	900

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: PPI subsistence and accommodation
Justification of Cost: 3x Subsistence for in-person workshops (£10/day x6 participants = £180 (£60 for each year of PhD)). 3x overnight accommodation for attendance of in-person workshop x6 participants (£100 per person = £1,800 total (£600 each year of PhD)). Total = £1980.

	Year 1	Year 2	Year 3	Total Cost (£)
	660	660	660	1980

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	8508	11565	6345	26418

Equipment
No costs have been added

Consumables

Description: GENEActiv accelerometer charger x12
Type of cost: HEI
Please describe fully: GENEActiv accelerometer charger x12
Justification of Cost: Accelerometers to measure physical activity levels during intervention period of WP2. A quote was received from ActivInsights company who produce the GENEActiv accelerometer and chargers. Prices for the chargers are £160.00 + VAT. 12 will be required for max participants per exercise group (12x £160 - £1,920 + VAT)

	Year 1	Year 2	Year 3	Total Cost (£)
	2320	0	0	2320

Description: GENEActiv accelerometer x12
Type of cost: HEI
Please describe fully: GENEActiv accelerometer x12
Justification of Cost: Accelerometers to measure physical activity levels during intervention period of WP2. A

quote was received from ActivInsights company who produce the GENEActiv accelerometer and chargers. Prices for the accelerometers are £250.00 + VAT. 12 will be required for max participants per exercise group (12x £250 - £3,000 + VAT)

	Year 1	Year 2	Year 3	Total Cost (£)
	3600	0	0	3600

Description: Research textbooks
Type of cost: HEI
Please describe fully: Research textbooks
Justification of Cost: Research textbooks (statistics, quantitative research methods, implementation science, designing clinical research) £100 each

	Year 1	Year 2	Year 3	Total Cost (£)
	400	0	0	400

Description: Encrypted audio recording device
Type of cost: HEI
Please describe fully: Encrypted audio recording device
Justification of Cost: Required for sound recording face to face qualitative interviews. Based on internet searches, cost = £450

	Year 1	Year 2	Year 3	Total Cost (£)
	450	0	0	450

Description: Exercise equipment for intervention during WP2
Type of cost: HEI
Please describe fully: Exercise equipment for intervention during WP2
Justification of Cost: Exercise equipment required for intervention in WP2 including theraband (£200), PD warrior equipment (£200), dumbbell weights (£200), exercise mats (£170), pilates blocks (£100), foam balls (£130)

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	0	0	1000

Description: Laptop for home working, remote working & fieldwork
Type of cost: HEI
Please describe fully: Laptop for home working, remote working & fieldwork
Justification of Cost: Laptop (excluding VAT) for home working, remote working & fieldwork. Approximately £1,000 based on internet searches

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	0	0	1000

Description: Laptop for use during WP2
Type of cost: HEI
Please describe fully: Laptop for use during WP2
Justification of Cost: Laptop (excluding VAT) for use during intervention in WP2 for instructors to use and to enable participants to join online. Based of current internet search approximately £1,000

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	0	0	1000

Description: Non-exercise equipment for WP2
Type of cost: HEI
Please describe fully: Non-exercise equipment for WP2
Justification of Cost: Non-exercise equipment required for intervention in WP2 and to complete outcome measures e.g. 2x stopwatch (£10/each = £20), tape measure (£10), music speakers (£80), pens (£10), clipboards x 12 (£30).

	Year 1	Year 2	Year 3	Total Cost (£)
	150	0	0	150

Description: Printing and postage during project and dissemination
Type of cost: HEI
Please describe fully: Printing and postage during project and dissemination
Justification of Cost: Printing and paper during years 1-3 for WP2 (consent form, PIS, outcome measures, questionnaires, participant exercise booklet) 2000 pages @5p = £100. Postage costs (recruitment materials, questionnaires, incentives) 200x £2 = £400. Consumables associated with dissemination. 2x conference poster printing = £100. Infographic printing = £200. Total printing and postage = £800+ VAT = £960.

	Year 1	Year 2	Year 3	Total Cost (£)
	240	360	360	960

Description: Technology equipment for WP2
Type of cost: HEI
Please describe fully: Technology equipment for WP2
Justification of Cost: Technology equipment for WP2 to run hybrid exercise intervention. Costs have been calculated from internet search for equipment: wide angle web camera (£90), tripod (£25), projector (£160), projector screen (£80), Ringlight (£80), wireless microphone system (£50), board and pens (£30).

	Year 1	Year 2	Year 3	Total Cost (£)
	515	0	0	515

Description: Tablet and dongles for WP2
Type of cost: HEI
Please describe fully: Tablet and dongles for WP2
Justification of Cost: Tablets and dongles for participants to use to participate in WP2 already available for use within Department of Clinical and Movement Neurosciences, Institute of Neurology.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

Description: Consumables (food and drink) for peer support session WP2
Type of cost: HEI
Please describe fully: Consumables (food and drink) for peer support session WP2
Justification of Cost: Consumables for peer support session WP2 e.g. tea, coffee, snacks, cups, plates etc.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	150	100	250

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	10675	510	460	11645

Patient and Public Involvement

Description: PPI payments
Type of cost: HEI
Please describe fully: PPI payments
Justification of Cost: All cost for PPI and advisory group involvement based on NIHR “Budgeting for involvement: Practical advice on budgeting for actively involving the public in research studies”. Over the 3 year project the following PPI will be included: PPI - 6x online meetings (2hrs x6 participants @ £50/h per person = £3,600 (£1,200 each year of PhD)). 3x in-person workshops (£300 p/day x6 participants = £5,400 (£1,800 each year of PhD)). Clinical academic advisory group: 8x online group meetings (1hr x5 participants @£150 each per meeting = £6,000 total over duration of PhD).

	Year 1	Year 2	Year 3	Total Cost (£)
	5250	5250	4500	15000

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	5250	5250	4500	15000

Other Direct Costs

Description: Venue hire
Type of cost: HEI
Please describe fully: Venue hire
Justification of Cost: Hiring of venue for delivery of intervention for WP2. Based on internet search of costs of hiring community venues. Venue hire (£120 p/hour x 3hrs = £360 x 64 weeks (duration of WP2 intervention in total = £23,040)

	Year 1	Year 2	Year 3	Total Cost (£)
	1440	14400	7200	23040

Description: Software licences for data analysis
Type of cost: HEI
Please describe fully: Software licences for data analysis
Justification of Cost: Software lisences for Microsoft Project (£300) & Office (£370), Stata (£250), Dragon (£450), Nvivo (£75)

	Year 1	Year 2	Year 3	Total Cost (£)
	1734	0	0	1734

Description: Transcription costs for WP2
Type of cost: HEI
Please describe fully: Transcription costs for WP2
Justification of Cost: Transcription of WP2 interviews (Transcription through professional provider costed at £0.97/min + VAT x 60min interviews x 50 participants = £3,492)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1746	1746	3492

Description: Incentives for participation
Type of cost: HEI
Please describe fully: Incentives for participation
Justification of Cost: Participation incentives for WP2. £10x 100 non-participants for TFA questionnaire = £1,000. £20x 50 for participation in WP2 intervention = £1,000. £10x 50 WP2 participant questionnaire & interview = £500. Total = £2,500.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1250	1250	2500

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	3174	17396	10196	30766

Training And Development

Type of cost: HEI
Description: Workshops
Item Description: Joanna Briggs Institute scoping review workshop
Justification of Cost: Course fees for JBI scoping review workshop based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	400	0	0	400

Type of cost: HEI
Description: Short Courses
Item Description: King's College London implementation science course
Justification of Cost: Course fees for KCL implementation science course based on previous fees

	Year 1	Year 2	Year 3	Total Cost (£)
	0	650	0	650

Type of cost: HEI
Description: Short Courses
Item Description: King's Fund Emerging Leaders course fees
Justification of Cost: Course fees for King's fund emerging leaders course based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	2700	2700

Type of cost: HEI
Description: Short Courses
Item Description: NCORE Atypical Parkinson's course
Justification of Cost: Course fees for NCORE atypical Parkinson's course based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	0	150	0	150

Type of cost: HEI
Description: Short Courses
Item Description: Oxford University Randomised Controlled Trials course
Justification of Cost: Course fees for Oxford Uni RCT course based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1500	0	1500

Type of cost: HEI
Description: Workshops
Item Description: PWR!Moves workshop
Justification of Cost: Course fees for PWR!Moves workshop based on current in-person fees

	Year 1	Year 2	Year 3	Total Cost (£)
	550	0	0	550

Type of cost: HEI
Description: Workshops
Item Description: Queen Mary University Pilot & Feasibility study workshop
Justification of Cost: Course fees for QMU pilot & feasibility study workshop based on previous fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	200	0	0	200

Type of cost: HEI
Description: Short Courses
Item Description: University of Salford Parkinson's disease course
Justification of Cost: Course fees for University of Salford Parkinson's disease module based on current fees

	Year 1	Year 2	Year 3	Total Cost (£)
	1500	0	0	1500

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: PhD fees
Justification of Cost: I am applying to undertake this PhD full time. UCL annual full time fees for 2024/25 are £6,040 per year. Total = £18,120. I understand NIHR will pay a maximum of £13,788

	Year 1	Year 2	Year 3	Total Cost (£)
	4596	4596	4596	13788

Type of cost: HEI
Description: Short Courses
Item Description: Parkinson's UK Train the Trainer course
Justification of Cost: Online Parkinson's UK train the trainer course - free attendance therefore no costs incurred

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	0	0

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	7246	6896	7296	21438

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	99512	105030	95501	300043

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: UCL
Justification of Cost: Estates costs for research as calculated by Imran Sayed and Clement Veall, Department of Clinical and Movement Neurosciences administrative officers. This figure includes infrastructure technicians.

	Year 1	Year 2	Year 3	Total Cost (£)
	32215	29530	26846	88591

Type of cost: Other Indirect Costs
Name of Higher Education Institute: UCL
Justification of Cost: Indirect costs associated with research programme as calculated by Imran Sayed and Clement Veall, Department of Clinical and Movement Neurosciences administrative officers

	Year 1	Year 2	Year 3	Total Cost (£)
	72904	66828	60753	200485

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	105119	96358	87599	289076

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	84095	77086	70079	231260

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	275605	231260	506865

Total Higher Education Institution Costs (100%)s	24438	0	24438
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	0	0	0
Total Research Costs Requested From Funder	300043	231260	531303

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**69981**	**70156**	**70804**	**210941**
Travel-subsistence-and-conference-fees	**7006**	**9453**	**5276**	**21735**
Equipment	**0**	**0**	**0**	**0**
Consumables	**8540**	**408**	**368**	**9316**
Patient-and-Public-Involvement	**4200**	**4200**	**3600**	**12000**
Other-Direct-Costs	**2539**	**13917**	**8157**	**24613**
Training And Development	**7246**	**6896**	**7296**	**21438**
HEI-Indirect-costs	**84095**	**77086**	**70079**	**231260**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**183607**	**182116**	**165580**	**531303**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
NHS service support cost from SoECAT	67	270	67	404.0

Justification:
NHS service support cost from SoECAT: NHS service support cost derived from SoECAT for WP2. This was completed in collaboration with the UCLH joint research office. Total = £404. I have distributed these as expected over the 3 years of the DCAF. Most NHS support costs will be incurred during WP2 during year 2.

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Support Costs Total	67	270	67	404

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
NHS treatment costs from SoECAT	0	0	0	0.0

Justification:

NHS treatment costs from SoECAT: NHS treatment costs derived from SoECAT. This was completed in collaboration with the UCLH joint research office. The intervention during WP2 being investigated is in addition to usual care therefore there are no excess NHS treatment costs.

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Treatment Costs Total	0	0	0	0

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
Yes

Usual Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
Usual treatment costs associated with WP2	0	0	0	0.0

Justification:
Usual treatment costs associated with WP2: There is no standardised usual care. The intervention during WP2 under investigation is in addition to usual care.

	Year 1	Year 2	Year 3	Total Cost (£)
Usual Treatment Costs Total	0	0	0	0

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	404
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	404

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	531303
Total NHS (Support and Excess Treatment) Costs	404
Total Funding Required	531707

10. Uploads

(References - DCAF references final.docx) is included as an appendix within this file.

(Figures / Tables - DCAF WP2 outcome measures table.docx) is included as an appendix within this file.

(Research Timetable - DCAF GANTT chart (2).docx) is included as an appendix within this file.

(SoECAT Template - Funder Export SoECAT Hybrid exercise for self-managed physical activity in PD.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Professor
Forename(s)	Anette
Surname	Schrag
Position	Professor

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Professor
Forename(s)	Fiona
Surname	Jones
Position	Professor Rehabilitation Research

Practice Supervisor(s)

Practice Support Participant 1

Title	Dr
Forename(s)	caroline
Surname	appel
Position	Clinical Lead Physiotherapist

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Dr
Forename(s)	caroline
Surname	appel
Position	Clinical Lead Physiotherapist

Head of Department or Senior Manager 2

Title	Professor
Forename(s)	Anthony
Surname	Schapira
Position	

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mr
Forename(s)	Dominic
Surname	Mikulski
Position	Assistant Director

12. Acknowledgement, Review and Submit

Conflict checks:
Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

There are no conflicts of interest to declare

Agreement to Terms and Conditions:

I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the NIHR research governance framework and the guidelines set out in 'Good Clinical Practice in MRC-funded Trials' (MRC 1998).

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Appendices

1. DCAF references final.docx
2. DCAF WP2 outcome measures table.docx
3. DCAF GANTT chart (2).docx
4. Funder Export SoECAT Hybrid exercise for self-managed physical activity in PD.xlsx

Appendix #1: DCAF references final.docx

(1) Dorsey ER, Elbaz A, Nichols E, Abbasi N, Abd-Allah F, Abdelalim A, et al. Global, regional, and national burden of Parkinson's disease, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. Lancet Neurology 2018 Nov 01,;17(11):939-953.
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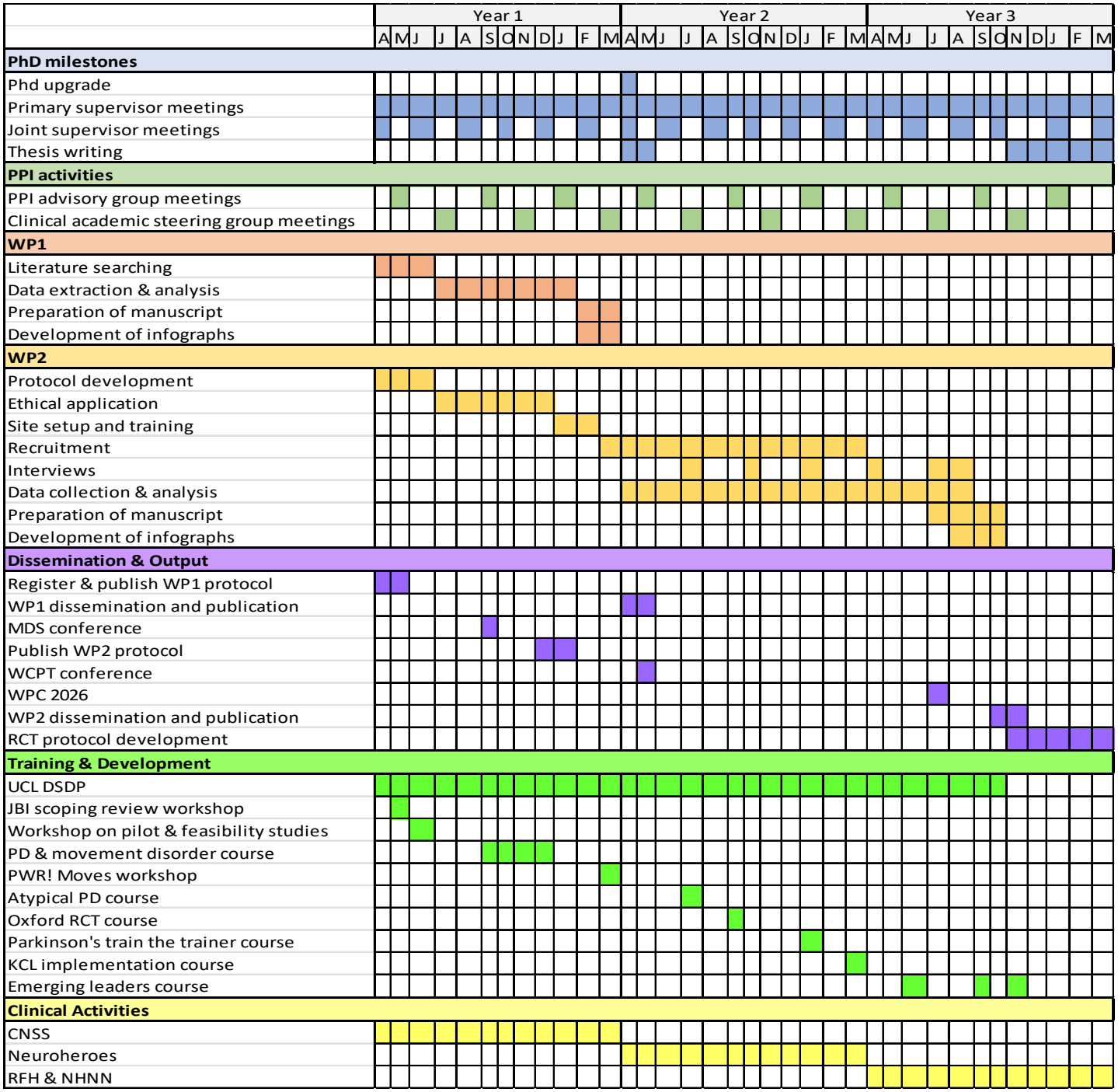
Appendix #2: DCAF WP2 outcome measures table.docx

Table 1. Outcome measures in relation to study aims and objectives.

Aim	Objective	Measures
Primary outcomes		
Acceptability	Representative recruitment and engagement Adherence to study	**Quantitative**: Demographic data (age, socioeconomic status, education level and ethnicity) Prospective acceptability through TFA questionnaire Adherence & attrition rates **Qualitative:** TFA questionnaire Interviews with participants and practitioners
Feasibility	Time to collect, clean and analyse data	Number of completed data sets Recruitment rates Retention rates
Fidelity	Practicality of carrying out intervention	**Quantitative:** Fidelity measure checklist **Qualitative:** Interviews with participants and practitioners Observation
Secondary outcomes		
Efficacy	Choice of outcome measures	Effect size calculations on: <u>Physical activity levels</u> Physical activity diaries Body worn accelerometer <u>Physical/motor function</u> Timed up and go (TUG) <u>Self-efficacy</u> Self-efficacy for exercise scale (SEE) Importance/Confidence scale

Appendix #3: DCAF GANTT chart (2).docx


Year 1
Year 2
Year 3
A M J J A S O N D J F M A M J J A S O N D J F M A M J J A S O N D J F M
PhD milestones
Phd upgrade
Primary supervisor meetings
Joint supervisor meetings
Thesis writing
PPI activities
PPI advisory group meetings
Clinical academic steering group meetings
WP1
Literature searching
Data extraction & analysis
Preparation of manuscript
Development of infographs
WP2
Protocol development
Ethical application
Site setup and training
Recruitment
Interviews
Data collection & analysis
Preparation of manuscript
Development of infographs
Dissemination & Output
Register & publish WP1 protocol
WP1 dissemination and publication
MDS conference
Publish WP2 protocol
WCPT conference
WPC 2026
WP2 dissemination and publication
RCT protocol development
Training & Development
UCL DSDP
JBI scoping review workshop
Workshop on pilot & feasibility studies
PD & movement disorder course
PWR! Moves workshop
Atypical PD course
Oxford RCT course
Parkinson's train the trainer course
KCL implementation course
Emerging leaders course
Clinical Activities
CNSS
Neuroheroes
RFH & NHNN

Appendix #4: Funder Export SoECAT Hybrid exercise for self-managed physical activity in PD.xlsx